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EXPOSURE APPARATUS, CONTROL METHOD THEREFOR, INFORMATION PROCESSING APPARATUS, INFORMATION PROCESSING METHOD, NON-TRANSITORY COMPUTER READABLE MEDIUM, AND ARTICLE MANUFACTURING METHOD

Abstract

An exposure apparatus for exposing an exposure region of a substrate by projecting a pattern of an original to the exposure region by a projection optical system, includes a measurement system configured to measure a height of the substrate at each of a plurality of measurement points in the exposure region, a user interface configured to prompt a user to make a setting concerning use of each of the plurality of measurement points, and a controller configured to control a focus operation of the exposure region with respect to an image plane of the projection optical system by using a measurement result of the measurement system based on the setting.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to an exposure apparatus, a control method therefor, an information processing apparatus, an information processing method, a non-transitory computer readable medium, and an article manufacturing method.

Description of the Related Art

[0002] As an exposure apparatus used in the manufacture of an article such as a semiconductor device, there are a step-and-repeat type exposure apparatus (stepper) and a step-and-scan type exposure apparatus (scanner). The stepper is a low cost apparatus as compared to the scanner, and is used in a step in which high resolution and high precision overlay are not required. In the stepper, after a substrate is step-driven so that a shot region on the substrate is positioned in an exposure region under a projection optical system, a height measurement apparatus measures the height, a focus operation is performed based on a measurement result, and then the shot region is exposed.

[0003] Japanese Patent Laid-Open No. 11-102852 discloses an exposure method of detecting the height of an exposure region on a substrate, calculating the average value of the highest height of the region and the lowest height of the region, and performing scan exposure while matching the image formation position of a projection optical system with the average height on the substrate.

[0004] In an exposure apparatus that measures the height of the substrate at each of a plurality of measurement points in an exposure region on the substrate and performs a focus operation based on the measurement results, it may be undesirable to perform the focus operation using the measurement results at all the measurement points. For example, if a focus operation is performed in consideration of a measurement result at a measurement point arranged in a region that has low repeatability among a plurality of exposure regions and is not important for the focus operation, an important region may be exposed in a defocus state.

SUMMARY OF THE INVENTION

[0005] The present invention provides a technique advantageous in adjusting a focus operation in an exposure apparatus including a measurement system capable of measuring the height of a substrate at each of a plurality of measurement points in an exposure region.

[0006] One of aspects of the present invention provides an exposure apparatus for exposing an exposure region of a substrate by projecting a pattern of an original to the exposure region by a projection optical system, comprising: a measurement system configured to measure a height of the substrate at each of a plurality of measurement points in the exposure region; a user interface configured to prompt a user to make a setting concerning use of each of the plurality of measurement points; and a controller configured to control a focus operation of the exposure region with respect to an image plane of the projection optical system by using a measurement result of the measurement system based on the setting.

[0007] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a view exemplifying the configuration of an exposure apparatus according to an embodiment;

[0009] FIG. 2 is a view exemplifying the configuration of a measurement system;

[0010] FIG. 3 is a view for explaining an example of the operation principle of the measurement system;

[0011] FIG. 4 is a view exemplifying the arrangement of a plurality of measurement points in a measurement region of the measurement system;

[0012] FIG. 5 is a view exemplifying a user interface screen provided to a user by a display device forming a user interface of a controller;

[0013] FIG. 6 is a flowchart exemplifying the operation of the exposure apparatus or a control apparatus associating with setting of the plurality of measurement points and a focus operation; and

[0014] FIG. 7 is a block diagram exemplifying the configuration of the controller, the control apparatus, or an information processing apparatus that controls the exposure apparatus.

DESCRIPTION OF THE EMBODIMENTS

[0015] Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed invention. Multiple features are described in the embodiments, but limitation is not made to an invention that requires all such features, and multiple such features may be combined as appropriate. Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

[0016] In the specification and the accompanying drawings, directions will be indicated on an XYZ coordinate system in which directions parallel to the surface of a substrate are defined as the X-Y plane. Directions parallel to the X-axis, the Y-axis, and the Z-axis of the XYZ coordinate system are the X direction, the Y direction, and the Z direction, respectively. A rotation about the X-axis, a rotation about the Y-axis, and a rotation about the Z-axis are OX, OY, and OZ, respectively.

[0017] FIG. 1 is a view showing the configuration of an exposure apparatus **100** according to an embodiment. The exposure apparatus **100** is one form of a lithography apparatus used in a photolithography step for manufacturing an article such as a semiconductor device or a display device. The exposure apparatus **100** exposes a substrate **108** by projecting the pattern of an original to the substrate **108** by a projection optical system **104**. The exposure apparatus **100** can expose the substrate **108** by a step-and-repeat method or a step-and-scan method. The present invention is not limited to the exposure method, but an example in which the exposure apparatus **100** is a step-and-repeat type exposure apparatus will be described below.

[0018] The exposure apparatus **100** exposes an exposure region (to be also referred to as a shot region hereinafter) of the substrate **108** by projecting the pattern of an original **103** to the exposure region by the projection optical system **104**. The substrate **108** has a photosensitive material such as a photoresist on its surface, and exposure of the substrate **108** indicates exposure of the photosensitive material. An illumination optical system **102** illuminates the original **103** by guiding light from a light source **101**. The light source **101** can be formed by, for example, an i-line mercury lamp, an excimer laser, or the like. A pattern to be projected is drawn on the original **103**. The light having passed through the original **103** reaches the substrate **108** on a stage **105** through the projection optical system **104** to form the image of the pattern of the original **103**. The stage **105** includes a substrate chuck that holds the substrate **108**. A substrate driving mechanism **110** can drive or position the stage **105** so that the substrate **108** is driven or positioned with respect to six axes including the X-axis, the Y-axis, the Z-axis, θ X-axis, the θ Y-axis, and the θ Z-axis. The image of the pattern of the original **103** projected on the substrate **108** is transferred to the photosensitive material such as a resist applied in advance to the surface of the substrate **108**. By repeating an operation of step-driving the stage **105** that holds the substrate **108** and an operation of exposing

the exposure region (shot region), the image of the pattern of the original **103** is transferred as a latent image to the photosensitive material in the plurality of exposure regions on the substrate **108**. A position measurement apparatus such as an interferometer or an encoder (not shown) measures the position and posture of the stage **105** with high accuracy, and driving or positioning of the stage **105** (substrate **108**) by the substrate driving mechanism **110** can be controlled in accordance with a measurement result. This can implement overlay exposure with high accuracy. The pattern transferred as a latent image to the photosensitive material in the exposure step is converted into a physical pattern in a development step.

[0019] In the exposure step, a measurement system **106** measures the height of the substrate **108** in the exposure region to adjust the height and tilt of the substrate **108** with respect to the image of the pattern of the original **103** (that is, the image plane of the projection optical system **104**). Then, based on a measurement result, at least one of the height and the tilt of the stage **105** can be controlled. Controlling at least one of the height and the tilt of the stage **105** to match the reference surface (for example, the surface or a surface offset from the surface) of the photosensitive material in the exposure region on the substrate **108** with the image plane of the projection optical system **104** will be referred to as a focus operation hereinafter.

[0020] The measurement system **106** can be configured to measure a height distribution that can be represented by the heights of the substrate **108** at a plurality of measurement points in the exposure region (shot region) on the substrate **108**. The size of the measurement region as the visual field of the measurement system **106** can be determined in accordance with required specifications. In an example, the measurement region of the measurement system **106** can be determined to cover the entire maximum exposure region in the exposure apparatus **100**. Alternatively, the measurement region may be a region smaller than the maximum exposure region in the exposure apparatus **100**. An example in which the measurement region of the measurement system **106** covers the entire exposure region (shot region) of the substrate **108**, that is, an example in which the height distribution of the exposure region can be obtained by performing measurement once by the measurement system **106** will be described below. However, the height distribution of the exposure region may be obtained by performing measurement a plurality of times by the measurement system **106**.

[0021] The configuration of the measurement system **106** will be described with reference to FIG. 2. The measurement system **106** detects the height at each of a plurality of measurement points in the measurement region. The measurement system **106** can include a light projector **201** and a light receiver **205**. The light projector **201** can include a light source **202** that makes light obliquely enter the surface of the substrate **108** as an object on the stage **105**, a two-dimensionally arrayed light projection pattern **203**, and a light projecting optical system **204** used to project the light projection pattern **203** to the object. Note that the light projecting optical system **204** can be omitted depending on the kind of the light source **202** and/or the distance between the light projection pattern **203** and the object. Alternatively, the light projection pattern **203** and the object may be set in the relationship of a Scheimpflug optical system by using the light projecting optical system **204**. By adopting the Scheimpflug optical system, the entire surface of the light projection pattern **203** can be focused on the object, thereby improving the measurement accuracy. Furthermore, when measuring the height of the object, the Scheimpflug optical system can prevent the measurement value from changing due to a local tilt of the object.

[0022] The light receiver **205** can include a light receiving optical system **207** and a camera **208**. The light reflected by the object enters the camera **208** via the light receiving optical system **207**. The camera **208** can include an image sensor **206** that includes a plurality of two-dimensionally arrayed pixels. Note that the light receiving optical system **207** can be omitted depending on the kind of the light source **202** and the distance between the object and the camera **208**. Alternatively, the object and the image sensor **206** may be set in the relationship of a Scheimpflug optical system by using the light receiving optical system **207**. By adopting the Scheimpflug optical system, the

entire surface of object can be focused on the image plane of the image sensor **206**. Furthermore, when measuring the height of the object, the Scheimpflug optical system can prevent the measurement value from changing due to the tilt of the object.

[0023] The exposure apparatus **100** includes a controller **107**. The controller **107** can be formed as, for example, a control apparatus that controls the operation of the exposure apparatus **100** by controlling the light source **101**, the illumination optical system **102**, the projection optical system **104**, the measurement system **106**, the substrate driving mechanism **110**, and the like. The controller **107** can be formed by a general-purpose or dedicated computer incorporating a program.

[0024] FIG. 7 shows an example of the configuration of the controller **107**. The controller **107** can include, for example, a CPU (processor) **810**, a memory **820**, a network interface **830**, a user interface **840**, and a device interface **850**. A program **822** to be executed by the CPU **810** can be held in or loaded into the memory **820**. The program **822** can be stored in a computer-readable memory medium and used for a transaction. For example, the program **822** stored in a computer-readable memory medium (not shown) can be read out from the memory medium by the device interface **850** and loaded into the memory **820**. Alternatively, the program **822** may be loaded from an external device holding it into the memory **820** via a network NW. The program **822** loaded into the memory **820** may be held non-transitory in the memory **820**. The user interface **840** includes, for example, a keyboard, a pointing device such as a mouse, and a display device (including a touch panel display), and can provide information to the user and acquire information from the user. The device interface **850** has a function of performing connection to various kinds of devices or components such as the light source **101**, the illumination optical system **102**, the projection optical system **104**, the measurement system **106**, and the substrate driving mechanism **110**.

[0025] FIG. 3 exemplarily shows a light projection pattern image (the image of the light projection pattern image **203**) **301** formed by projecting the light projection pattern to the substrate **108** by the light projector **201** of the measurement system **106**. The light projection pattern image **301** can be formed in a measurement region **302** of the measurement system **106**. The light projection pattern image **301** can include a plurality of bars, as schematically shown in FIG. 3. To precisely measure a height map (height distribution) in the measurement region **302** (or the exposure region), it is necessary to reduce the interval between the bars in the light projection pattern image **301**. If the image sensor **206** is a two-dimensional image sensor, a height map in the measurement region **302** of the surface of the substrate **108** can be obtained at a measurement interval defined by the light projection pattern image **301** projected to the substrate **108** and the pixel density of the image sensor **206**.

[0026] The light projection pattern image **301** has information corresponding to the height distribution of the surface of the substrate **108** in the measurement region **302** (exposure region). The controller **107** can obtain the height map (height distribution) in the measurement region **302** (exposure region) based on the light projection pattern image **301** captured by the image sensor **206**. Note that the controller **107** may execute at least part of processing of forming the height map (height distribution) based on the output from the image sensor **206**. In this case as well, the measurement system **106** can be considered to measure the height in the measurement region **302** (exposure region). The measurement system **106** can measure the height of the surface of the substrate **108** at each of a plurality of measurement points in the measurement region **302** (exposure region). The controller **107** can control the focus operation of the exposure region with respect to the image plane of the projection optical system **104** based on the output from the measurement system **106**.

[0027] FIG. 4 shows an example of the arrangement of a plurality of measurement points **401** in the measurement region **302**. The light projection pattern image **301** exemplified in FIG. 3 may be formed (projected) at each measurement point **401** or formed (projected) over the measurement region **302**. The plurality of measurement points **401** are uniformly arranged in the measurement region **302** in the example shown in FIG. 4, but the plurality of measurement points **401** need not

uniformly be arranged. The controller **107** can obtain a height map in the exposure region on the surface using the measurement system **106** before setting of the focus operation to be described below, and the height map can be provided to the user by the display device forming the user interface **840**.

[0028] FIG. **5** exemplarily shows a user interface screen **500** provided to the user by the display device forming the user interface **840** of the controller **107**. The display format, the display position, and the like of each piece of information that can be provided to the user by the user interface screen **500** can be changed appropriately.

[0029] A display region **501** is a region where a shot layout representing the arrangement of a plurality of shot regions (exposure regions) defined on the substrate **108** is displayed. A shot region indicated by dotted lines represents a shot region for editing, and in the example shown in FIG. **5**, a shot region **S1** is selected for editing. Shot regions indicated by alternate long and short dashed lines represent a plurality of sample shot regions for evaluating the repeatability of a height measurement result. The plurality of sample shot regions can arbitrarily be set by the user. The step-and-repeat type exposure apparatus sequentially executes exposure processing for a plurality of shot regions **S1** to **S32** by stepping. An example in which the measurement region **302** is the same as the exposure region (shot region) or covers the entire exposure region (shot region) will be described.

[0030] A display region **502** is a region where parameters are displayed. More specifically, the display region **502** is a region for displaying parameters associated with pieces of information to be displayed in a plurality of display regions **503**, **504**, and **505** included in the user interface screen **500**. A parameter **502a** represents a threshold for evaluating the repeatability of a height measurement result. In this example, if a numerical value indicating the repeatability of a height measurement result among the plurality of sample shot regions (selected exposure regions) is smaller than the threshold set as the parameter **502a**, it is determined that the repeatability is high, and if the numerical value exceeds the threshold, it is determined that the repeatability is low. The repeatability is evaluated for the measurement result at each of the plurality of measurement points. The repeatability is an index (index value) representing variations among the plurality of sample shot regions (selected exposure regions), and can be, for example, a standard deviation (σ).

[0031] Parameters **502b** and **502c** are parameters for setting the number of dies forming the shot region (exposure region). The parameter **502b** provides the user with a function of setting the number of dies in the X direction (the division number in the X direction), and the parameter **502c** provides the user with a function of setting the number of dies in the Y direction (the division number in the Y direction).

[0032] The display region **503** is a region where process information is displayed. The process information can include a height map **503a** of the shot region (exposure region) of the substrate and die information **503b**. In the example shown in FIG. **5**, the height map **503a** is the height map of the shot region **S1** selected in the display region **501**, and the die information **503b** indicates the arrangement of the plurality of dies in each shot region in the shot layout shown in the display region **501**. Note that the display example shown in FIG. **5** is merely an example and, for example, the height map **503a** may display the height by color, may be changeable in scale, or may display the difference of the height map of another shot region. The die information **503b** is displayed in accordance with the pieces of input information to the parameters **502b** and **502c** of the display region **502**. The parameter **502b** is a parameter for setting the number of dies in the X direction (the division number in the X direction), and the parameter **502c** is a parameter for setting the number of dies in the Y direction (the division number in the Y direction).

[0033] The display region **504** is a region where information concerning the measurement system **106** (focus sensor) is displayed. The display region **504** can include a repeatability map **504a** obtained by mapping repeatability information representing the repeatability of the height measurement result to each of the plurality of measurement points **401**, and a setting map **504b**

obtained by mapping a setting state associated with the use of each of the plurality of measurement points **401** of the measurement system **106** to each of the plurality of measurement points **401**. [0034] The repeatability map **504a** indicates a result of evaluating the repeatability of each of the height measurement results obtained by measuring in advance, by the measurement system **106**, the sample shot regions **S6**, **S9**, **S11**, **S17**, **S24**, and **S27** set in the display region **501**. The sample shot regions can be changed by the user. In this evaluation, if a numerical value representing the repeatability of the height measurement result among the plurality of sample shot regions is smaller than the threshold set as the parameter **502a**, it is determined that the repeatability is high, and if the numerical value exceeds the threshold, it is determined that the repeatability is low. In the display region **504**, the measurement point **401** at which the numerical value representing the repeatability is smaller than the threshold and the measurement point **401** at which the numerical value representing the repeatability exceeds the threshold can be distinguished and displayed. In the example shown in FIG. 5, the measurement point (Low Repeatability) at which the numerical value representing the repeatability is smaller than the threshold is indicated by “x” and the measurement point (High Repeatability) at which the numerical value representing the repeatability exceeds the threshold is indicated by “O”. Note that a method of evaluating the repeatability and a method of presenting the evaluation result are not limited to the above example. The repeatability map **504a** as information displayed in the display region **504** may be understood to include information representing the arrangement of the plurality of measurement points **401** arranged in the measurement region or the exposure region.

[0035] The setting map **504b** provides a user interface for prompting the user to make a setting concerning the use of each of the plurality of measurement points **401** in the focus operation. In an example, the setting map **504b** provides a user interface for prompting the user to set whether to use each of the plurality of measurement points **401** in the focus operation. In the example shown in FIG. 5, the measurement point **401** (Valid Sensor) used for the focus operation is indicated by “O”, and the measurement point **401** (Invalid Sensor) not used for the focus operation is indicated by “-”. Distinguishment between the measurement point used for the focus operation and that not used for the focus operation is not limited to this, and for example, these measurement points may be distinguished by colors or the like. The setting map **504b** as information displayed in the display region **504** may be understood to include information representing the arrangement of the plurality of measurement points **401** arranged in the measurement region or the exposure region.

[0036] The measurement point **401** (Valid Sensor) used for the focus operation is the measurement point **401** used by the controller **107** to control the focus operation, and the controller **107** controls the focus operation based on the measurement result at the measurement point **401**. The measurement point **401** (Invalid Sensor) not used for the focus operation is the measurement point **401** not used by the controller **107** to control the focus operation, and the controller **107** ignores the measurement result at the measurement point **401** in the control of the focus operation.

Alternatively, with respect to the measurement point **401** not used for the focus operation, the measurement system **106** need not perform measurement, need not output the measurement result, or may output an invalid value. For example, the user can double-tap or click, by the pointing device, the measurement point (in FIG. 5, the position at which “O” or “-” is displayed) in the setting map **504b**, thereby setting whether to use the measurement point. In another example, the setting map **504b** can provide a user interface for prompting the user to set a weight for the measurement result at each of the plurality of measurement points **401**. In this case, the controller **107** can set, for the measurement result at each of the plurality of measurement points **401**, a weight corresponding to it in the focus operation, and execute the focus operation based on the weighted measurement results.

[0037] The display region **505** can display at least two of information representing the arrangement of the plurality of measurement points, information representing the height map in the exposure region, information representing the arrangement of the plurality of dies in the exposure region, and

information representing the evaluation result of the repeatability of the measurement result at each of the plurality of measurement points. The display region **505** can be configured to superimpose and display at least two of these pieces of information.

[0038] The user interface **840** or the user interface screen **500** exemplified in FIG. 5 is effective for the user to make a setting concerning the use of each of the plurality of measurement points **401**. For example, the user can make a setting concerning the use of each of the plurality of measurement points based on the pieces of information of the arrangement of the plurality of measurement points, the height map, the arrangement of the plurality of dies, the repeatability of the measurement result, and the like. For example, the user can set, as a measurement point not to be used, the measurement point located at the peak or valley in the height map based on the height map. Alternatively, the user can set the measurement point having low repeatability as a measurement point not to be used.

[0039] FIG. 6 exemplarily shows the operation of the exposure apparatus **100** associated with setting of the plurality of measurement points and the focus operation. This operation can be controlled by the controller **107**, more specifically, the CPU (processor) **810** that operates in accordance with the program **822**.

[0040] In step **S601**, the controller **107** acquires height map information representing the height distribution of the exposure region (shot region) of the substrate **108**. In an example, the controller **107** can acquire the height map information in the exposure region (shot region) based on the output from the measurement system **106**. The height map information is obtained by, for example, measuring, by the measurement system **106**, the height distributions of the plurality of exposure regions (shot regions) of the substrate **108** loaded to the exposure apparatus **100** and held by the stage **105**. In another example, the controller **107** can acquire, from the memory **820**, height map information associated in advance with a lot including a plurality of substrates **108** to undergo the exposure operation in the exposure apparatus **100**. Alternatively, the controller **107** may acquire, from another apparatus (for example, a management apparatus) connected to the network interface **830**, height map information associated in advance with the lot including the plurality of substrates **108** to undergo the exposure operation in the exposure apparatus **100**.

[0041] In step **S602**, the controller **107** can generate repeatability information representing the repeatability of the height measurement result based on the height map information acquired in step **S601**. For example, the controller **107** can generate repeatability information by calculating an index (index value) indicating variations among the plurality of sample shot regions based on the height map information. In another example, the controller **107** can acquire, from the memory **820**, repeatability information associated in advance with the lot including the plurality of substrates **108** to undergo the exposure operation in the exposure apparatus **100**. Alternatively, the controller **107** may acquire, from another apparatus (for example, a management apparatus) connected to the network interface **830**, repeatability information associated in advance with the lot including the plurality of substrates **108** to undergo the exposure operation in the exposure apparatus **100**.

[0042] In step **S603**, the controller **107** displays, on the user interface (UI) **840**, the height map **503a** corresponding to the height map information acquired in step **S601**. In step **S604**, the controller **107** displays, on the user interface (UI) **840**, the repeatability map **504a** obtained by mapping the repeatability information acquired or generated in step **S602** to the plurality of measurement points.

[0043] In step **S605**, the controller **107** displays the setting map **504b** on the user interface (UI) **840**, and prompts the user to set whether to use each of the plurality of measurement points **401** in the focus operation.

[0044] In step **S606**, the controller **107** controls the focus operation of the exposure region with respect to the image plane of the projection optical system **104** by using the measurement result of the measurement system **106** based on the setting made in step **S605**, and exposes each exposure region of the substrate **108**. Step **S606** can be executed for, for example, the lot including the

plurality of substrates.

[0045] The controller **107** may be formed as an information processing apparatus for controlling the focus operation in the exposure apparatus **100** including the measurement system capable of measuring the height of the substrate **108** at each of the plurality of measurement points **401** in the exposure region of the substrate **108**. The controller **107** that is formed as such an information processing apparatus may or may not be connected to the exposure apparatus **100**. The controller **107** or the CPU **810** (processor) can include the user interface **840** for prompting the user to make a setting concerning the use of each of the plurality of measurement points **401** in the exposure region. This setting can designate that the exposure apparatus **100** performs the focus operation by using the measurement result of the measurement system **106** based on the setting.

[0046] An article manufacturing method of manufacturing an article using the exposure apparatus **100** will be described below. The article manufacturing method can include an exposure step of exposing the substrate **108** using the exposure apparatus **100**, a development step of developing the substrate **108** having undergone the exposure step, and a step of obtaining an article from the substrate **108** having undergone the development step. A photosensitive material (photoresist) is applied to the substrate **108** provided to the exposure apparatus **100**. In the exposure step, the pattern of the original is transferred as a latent image pattern to the photosensitive material. In the development step, the latent image pattern is converted into a physical device pattern. The step of obtaining the article from the substrate **108** having undergone the development step can include, for example, a step of patterning the underlying layer using the device pattern. Furthermore, the step of obtaining the article from the substrate **108** having undergone the development step may include a step of dicing the substrate.

OTHER EMBODIMENTS

[0047] Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

[0048] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0049] This application claims the benefit of Japanese Patent Application No. 2024-027869, filed Feb. 27, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

- 1.** An exposure apparatus for exposing an exposure region of a substrate by projecting a pattern of an original to the exposure region by a projection optical system, comprising: a measurement system configured to measure a height of the substrate at each of a plurality of measurement points in the exposure region; a user interface configured to prompt a user to make a setting concerning use of each of the plurality of measurement points; and a controller configured to control a focus operation of the exposure region with respect to an image plane of the projection optical system by using a measurement result of the measurement system based on the setting.
- 2.** The apparatus according to claim 1, wherein the user interface prompts the user to set whether to use each of the plurality of measurement points.
- 3.** The apparatus according to claim 1, wherein the user interface prompts the user to set a weight for the measurement result at each of the plurality of measurement points, and the controller controls the focus operation based on the weight set for the measurement result at each of the plurality of measurement points.
- 4.** The apparatus according to claim 1, wherein the user interface displays information representing an arrangement of the plurality of measurement points.
- 5.** The apparatus according to claim 4, wherein the user interface further displays information representing a height map in the exposure region.
- 6.** The apparatus according to claim 4, wherein the user interface further displays information representing an arrangement of a plurality of dies in the exposure region.
- 7.** The apparatus according to claim 4, wherein the user interface further displays information representing an evaluation result of repeatability of the measurement result at each of the plurality of measurement points.
- 8.** The apparatus according to claim 7, wherein the repeatability indicates repeatability among a plurality of exposure regions of the substrate.
- 9.** The apparatus according to claim 7, wherein the repeatability is an index indicating variations among a plurality of exposure regions of the substrate.
- 10.** The apparatus according to claim 1, wherein the user interface displays at least two of information representing an arrangement of the plurality of measurement points, information representing a height map in the exposure region, information representing an arrangement of a plurality of dies in the exposure region, and information representing an evaluation result of repeatability of the measurement result at each of the plurality of measurement points.
- 11.** The apparatus according to claim 1, wherein the user interface superimposes and displays at least two of information representing an arrangement of the plurality of measurement points, information representing a height map in the exposure region, information representing an arrangement of a plurality of dies in the exposure region, and information representing an evaluation result of repeatability of the measurement result at each of the plurality of measurement points.
- 12.** An information processing apparatus for controlling a focus operation in an exposure apparatus including a measurement system capable of measuring a height of a substrate at each of a plurality of measurement points in an exposure region of the substrate, comprising: a user interface configured to prompt a user to make a setting concerning use of each of the plurality of measurement points in the exposure region, wherein the setting designates that the exposure apparatus performs the focus operation by using a measurement result of the measurement system based on the setting.
- 13.** A control method for an exposure apparatus including a measurement system capable of measuring a height of a substrate at each of a plurality of measurement points in an exposure region of the substrate, comprising: prompting a user to make a setting concerning use of each of the plurality of measurement points in the exposure region; and exposing the exposure region by controlling a focus operation of the exposure region with respect to an image plane of a projection

optical system of the exposure apparatus by using a measurement result of the measurement system based on the setting.

14. A non-transitory computer readable medium storing program for causing a computer to execute a control method for an exposure apparatus defined in claim 13.

15. An information processing method of controlling a focus operation in an exposure apparatus including a measurement system capable of measuring a height of a substrate at each of a plurality of measurement points in an exposure region of the substrate, comprising: prompting a user to make a setting concerning use of each of the plurality of measurement points in the exposure region, wherein the setting designates that the exposure apparatus performs the focus operation by using a measurement result of the measurement system based on the setting.

16. A non-transitory computer readable medium storing program for causing a computer to execute an information processing method defined in claim 15.

17. An article manufacturing method comprising: exposing a substrate using an exposure apparatus defined in claim 1; developing the substrate having undergone the exposing; and obtaining an article by processing the substrate having undergone the developing.

18. An article manufacturing method comprising: exposing a substrate by controlling an exposure apparatus in accordance with a control method for the exposure apparatus defined in claim 13; developing the substrate having undergone the exposing; and obtaining an article by processing the substrate having undergone the developing.
